F-22E Strike Raptor FCS: How the Control Laws Work

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This guide explains, axis by axis and surface by surface, how the fly-by-wire (FBW) flight control system in the F-22E Strike Raptor FCS addon (version 1.9.7) decides what every control surface and the thrust-vectoring nozzles do. It is written for someone who likes jets but has never written a control law. Each section starts with the plain idea, then gives the real numbers the addon uses.

The addon runs next to Aryx's unmodified F-22E Strike Raptor mod (1.0.4) in Nuclear Option as a BepInEx plugin. It replaces how the jet is flown, not how it looks or how the game's physics work.

Words you will see a lot

Term	What it means in plain words
AoA (angle of attack, α)	The angle between where the nose points and where the jet is actually going. More AoA = more lift, until the wing stalls.
G (load factor)	How hard the jet is turning, as a multiple of gravity. 9 G = the pilot feels nine times heavier.
IAS (indicated airspeed)	Speed as the wings "feel" it. Control power grows with IAS; at low IAS every surface is weak.
Dynamic pressure ($\bar{q}$)	The air's push on the jet. Roughly speed squared times air density. It is what IAS really measures.
Pitch / roll / yaw rate	How fast the jet rotates nose-up/down, around its long axis, and nose-left/right, in degrees per second (°/s).
Sideslip (β)	How much the air hits the side of the jet. Zero sideslip = flying "clean", nose into the wind.

Term	What it means in plain words
Velocity-vector roll	Rolling around the direction the jet is travelling, not around its own nose. This is what keeps a high-AoA roll from turning into a sideways slide.
Stabilators ("stabs")	The two all-moving horizontal tails. Moved together they pitch the jet; moved opposite they roll it.
Flaperons	The trailing-edge surfaces on the wings. The F-22E has two pairs (inner and outer). They act as flaps and as ailerons.
Rudders	The moving parts of the two canted (tilted-out) vertical tails.
TVC (thrust vectoring)	The engine nozzles pointing up or down to push the tail. On this jet it is pitch-only, like the real F-22.
Authority	How much rotation a surface can produce right now. It changes with speed, AoA and whether the surface is stalled.
Departure	Losing control: the jet goes somewhere the pilot did not ask for (spin, tumble, runaway AoA).
MPO (manual pitch override)	The mode that turns the AoA limit off so the pilot can go past it on purpose.
Bench	The 6-degree-of-freedom simulator the addon was tuned on, using the real F-22E airframe data extracted from Aryx's mod.

The big picture

The pilot's stick and pedals no longer move surfaces directly. They ask for a result (an AoA, a G, a roll rate, a sideslip) and the FCS works out, on every physics tick of the game, which mix of surfaces and nozzles delivers it without losing control.

What the stock mod did, and why it needed replacing

- **No AoA protection.** Nothing stopped the jet pulling into a departure.

- **No real AoA or G command.** The stick was closer to "move the tail" than "give me this turn".
- **Roll and yaw as separate axes.** At high AoA that makes a roll turn into a sideslip, because rolling around the nose is not rolling around the flight path.
- **A fixed roll-rate target.** The stock roll law aimed at a constant 286 °/s and used roll thrust vectoring, which the real F-22 does not have.

The design rules the addon follows

1. **Command the result, not the surface.** Stick pitch = AoA or G (or pitch rate). Stick roll = roll rate about the flight path. Pedal = sideslip.
2. **Measure, don't guess.** Wherever possible the FCS works from what the jet is doing right now (measured rotation, live thrust, live mass) rather than from speed-based lookup tables ("gain scheduling"). A heavy store or battle damage then shows up automatically.
3. **Know every surface's real aerodynamics.** The FCS carries a model of each surface built from the same parts, airfoils and lift curves the game uses. It knows when a surface is stalled, when it has reversed, and how much pitch, roll and yaw each degree of deflection really gives.
4. **Never command what the jet can't stop.** Every rate limit is set by how fast the jet could brake that rate again with the authority it has now.
5. **Released stick = hold still.** Letting go holds zero pitch rate and zero roll rate. The jet stays where you left it.
6. **Smooth beats snappy when they conflict.** Small commands build gently, big ones build fast, and nothing overshoots its limit.

The toolbox: what the FCS can move

The FCS flies nine things: two stabilators, four flaperons, two rudders and the pitch nozzles. Every one of them moves at 100 °/s with a 0.06 s ramp-up, so nothing can jump instantly and the FCS has to plan ahead.

Effector	Travel	Full swing time	Main jobs	Weak spot
Stabilators (left + right)	+35° trailing edge up / −30° trailing edge down	0.71 s	Pitch (together), roll (opposite), some yaw at high AoA	At moderate AoA the one going trailing-edge-down stalls first
Inner flaperons	±25°	0.56 s	Roll, flaps (extra lift), ground braking	Lose roll power past the wing stall
Outer flaperons	±25°	0.56 s	Roll, droop as flaps (60 % of the inner flap angle)	Same as inner
Rudders	±35°	0.76 s	Yaw, sideslip control, coordinating rolls	The fins are canted, so every rudder input also rolls the jet the opposite way
TVC nozzles (pitch only)	±20°	0.46 s	Pitch, especially at low speed and high AoA	Only as strong as the thrust; the game switches them off above 450 m/s true airspeed
Slats (leading edge)	Up / down	—	More wing area at AoA	Only effect in the game is added wing area
Speedbrake (split surfaces)	In / out	—	Drag	Manual only

Airframe tweaks shipped with the addon

Both the stock-performance and the buffed profile carry the same aero changes. These change how strong each surface is, not how far it moves.

- Flaperon effectiveness ×1.5.
- Stabilator effectiveness ×1.1.

- LERX (leading-edge root extension) ×1.6.
- Centre of gravity 0.05 m further aft (a little less stable, a little more agile; the FCS makes up the stability).

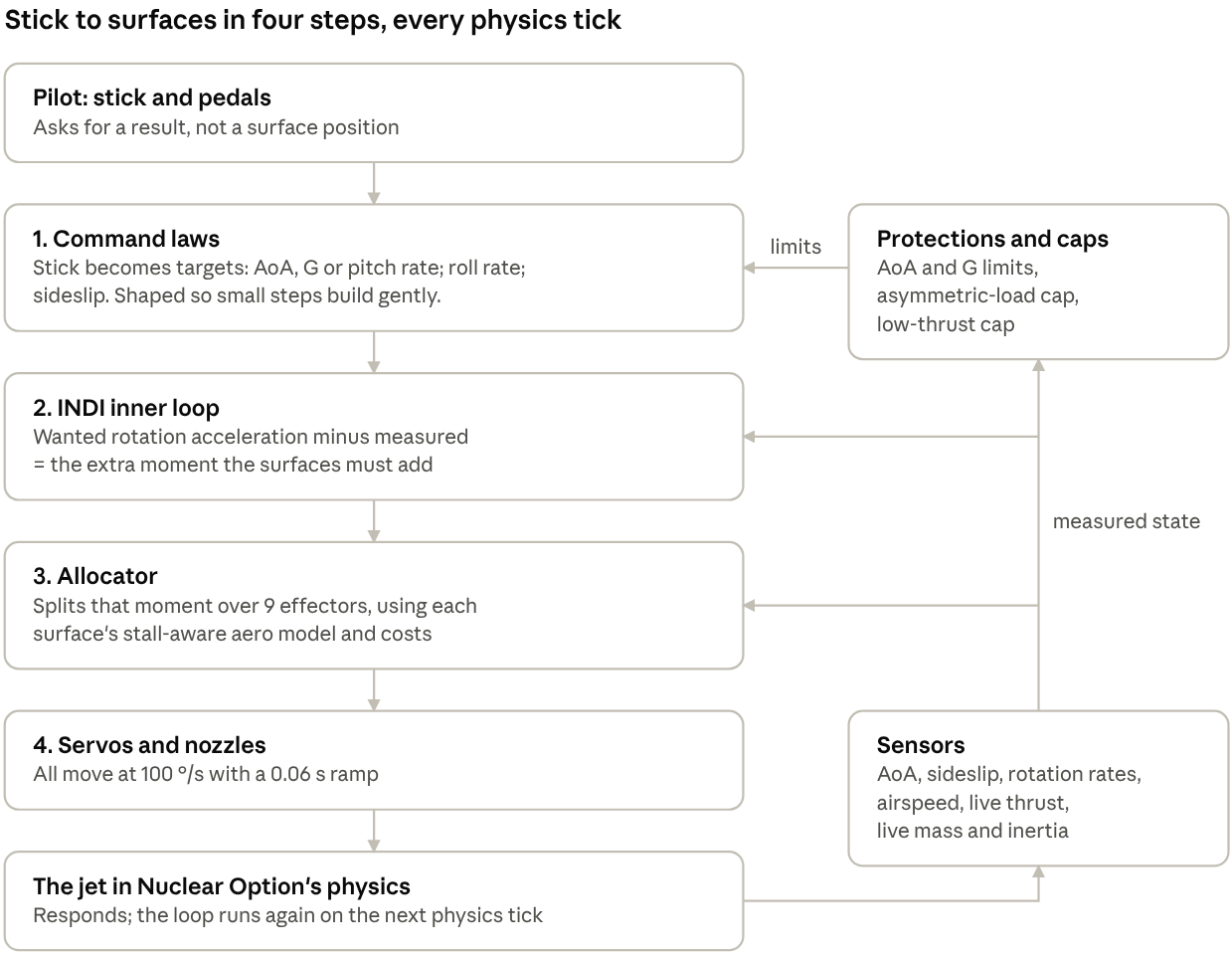
The buffed profile also adds thrust ×1.08, parasitic drag ×0.885, fuel burn ×0.75, and more countermeasures and gun ammo. The control laws are identical in both profiles, because the FCS reads the engines' live thrust every frame.

Why TVC is pitch-only, and what that costs

The real F-22 vectors its nozzles only up and down, so the addon does the same. The nozzles take pitch work off the stabs, which frees the stabs to help with roll and yaw. The price is that at very low speed there is no thrust-powered roll: full wing deflection gives only about 1.3 °/s of roll rate per m/s of true airspeed there (about 130 °/s at 100 m/s).

The brain: from stick to surfaces

On every physics tick the FCS runs the same four steps: turn the stick into targets, work out how much extra rotation is needed, split that across the surfaces, and move them. The whole loop is fed by what the jet is measured to be doing right now. It is tied to the game's physics step, not to a clock of its own: that is 50 times a second if Nuclear Option keeps Unity's default 0.02 s step, which the bench uses but which has not been confirmed in game.



FCS loop · command laws, INDI, allocator, servos, sensors

Step 2 in plain words: INDI

INDI stands for **Incremental Nonlinear Dynamic Inversion**. The idea is simple:

1. The command laws say: "I want the nose to accelerate upward at 30 °/s²."
2. The sensors say: "It is accelerating at 22 °/s² right now."
3. INDI asks the surfaces for **only the missing 8 °/s²**, on top of what they are already doing.

Because it works on the difference, INDI does not need a perfect model of the whole aircraft. Anything it cannot model (a store, damage, a gust, a weird mass distribution) is already inside the measured acceleration, so it gets corrected automatically. That is why the addon needs so little gain scheduling.

The pitch loop is a **hybrid** INDI. The sensors are filtered, which adds a few hundredths of a second of lag. At very high speed the airframe's own pitch moment changes fast enough that this lag made the G swing above and below its target. So the FCS predicts the lag from the airframe model (how pitch moment changes with AoA and pitch rate) and adds it back. This correction fades out from 25° to 40° AoA, where it is not needed.

Step 3 in plain words: the allocator

The allocator is like a manager handing out a job. It knows, for every surface at this exact AoA and speed, how much pitch, roll and yaw one more degree of deflection would really make, including whether that surface is stalled or even working backwards. Then it finds the combination that:

- **delivers the requested pitch, roll and yaw acceleration** as closely as possible;
- **prefers surfaces near neutral** and avoids big, fast swings;
- **never commands past a surface's travel** or faster than its servo can move;
- **follows extra rules,** such as keeping stab authority for roll when lateral stick is in, not letting groups of surfaces fight each other for drag, and giving the released-stick pitch hold priority.

One subtle fix matters at low speed: the "prefer neutral" preference is scaled down with dynamic pressure below about 150 m/s IAS. Without it, the allocator became lazy at low speed, because the moments it was chasing shrink with speed squared while its preferences did not. Removing that imbalance fixed weak low-speed roll and slow reversals.

Flight modes

The FCS picks its pitch law from three things: gear up or down, Stability Assist on or off, and whether a human or the AI is flying. Roll and yaw work the same way in every airborne mode.

Mode	When	Pitch stick means	Limits
Normal (human)	Gear up, Stability Assist on	AoA at low speed, G at high speed, pitch rate past about 36° AoA	+65° / −45° AoA, +9.5 / −3.5 G

Mode	When	Pitch stick means	Limits
Gear down	Landing gear down, airborne	Pitch rate (for precise approach flying)	Same protections
MPO	Stability Assist off	Pure pitch rate, up to the 48 °/s budget	No AoA limit at low speed; +12.5 / −3.5 G at high speed
AI jets	Any AI-flown F-22E	Pitch rate (the AI's autopilot expects rate)	Full FCS: same allocation, protection and lateral laws
Ground	Wheels on the runway	Direct stick-to-surface below 45 m/s (162 km/h)	Nose-wheel steering and aero-braking laws (see On the ground)
Rotation	Main wheels down, nose wheel up, above 45 m/s	Closed-loop pitch, so rotation and flare are smooth	Arms after the nose wheel is off for 0.15 s; disarms 8 m/s below 45 m/s

Switching without jolts

- **Debounces.** Ground-to-air needs 0.2 s with no wheel on the ground; a bumpy take-off roll can't flick the jet in and out of modes.
- **Protection override latch.** If you pull past the AoA limit in MPO and then turn Stability Assist back on while still holding full aft stick, the FCS does not snap the nose down. It holds zero pitch rate until you ease the stick below 90 %, then hands back to the normal law. It only arms if the jet is at least 5° past the limit.
- **Tail slides.** AoA is tracked "unwrapped" through ±180°, so a tail slide can't make the FCS think it has suddenly jumped to the negative limit.

What a player can change

The normal config shows only a few settings: the performance profile (stock or buffed, stock by default), Enabled, stick smoothing, the AI toggles, and per-profile flare, gun, EW capacity and EW recharge. Every tuning value in this guide lives in a separate developer build.

Pitch axis (stick forward and back)

In normal flight, pulling the stick asks for an **AoA** when the jet is slow and a **G** when it is fast, and past about 36° AoA it asks for a **pitch rate**. The FCS blends between these so you never feel a seam.

Step 1: how much pitch rate is allowed (the "rate budget")

Before anything else, the FCS decides how fast the nose may move right now. Everything else in pitch lives inside this budget.

- **Hard ceiling: 48 °/s.**
- **At low AoA** the budget follows airspeed: the faster you go, the more rate the wing can support.
- **At higher AoA** it follows thrust-to-weight, because there the engines (through the nozzles) do much of the work of holding the nose.
- **It uses live mass.** A heavy jet gets a slightly smaller budget automatically.

This logic was ported from Tom's SimplePlanes F-22 FCS.

Step 2: AoA or G? (the blend)

At low speed, even full aft stick cannot reach 9.5 G; the wing stalls first. So the stick asks for AoA there. At high speed, a small AoA already makes a huge G, so the stick asks for G. The FCS blends the two by asking one question: **within the rate budget at this speed, which limit would the jet hit first, the AoA limit or the G limit?** The answer slides smoothly with speed, so there is no switch point.

	Positive (pull)	Negative (push)
AoA limit	+65°	−45°
G limit	+9.5 G	−3.5 G
MPO G limit	+12.5 G	−3.5 G

Step 3: the stick map (low speed, AoA side)

The stick has a **fixed scale of 72° of AoA per full stick**, so **half stick = 36°**, and holding half stick holds 36°.

- **Below the hand-over threshold** (normally 36°), the stick is a pure AoA command. Stick position = AoA, one-to-one.
- **Full aft stick** acts like asking for 65°. The jet swings through 36° at the full rate budget with no pause at the threshold, then is caught at the limit.
- **Past the threshold the stick becomes a pitch-rate command** (next step).

Why the hand-over? Past about 36° the wing is past its stall. "Hold 50° AoA" stops being useful to a pilot; "move the nose this fast" is what you actually want when pointing the nose at a target.

Step 4: the hand-over to pitch-rate command (above ~36°)

The threshold moves with speed. It is 36° while the wings alone can still lift at least 2× the jet's weight. As that falls to 1× weight, the threshold drops linearly to 25°. On the bench that is roughly full 36° above ~330 km/h and 25° at ~210 km/h. It is smoothed over 0.5 s so it never jitters. The stick point of the threshold moves with it (for example 24° sits at 0.333 stick).

What happens when the jet crosses the threshold with the stick past its point:

1. **No jump.** The stick becomes a rate command starting from whatever pitch rate the jet had at the crossing.
2. **Pull further** and the rate rises toward the 48 °/s budget at full stick. **Ease off** and the rate falls linearly toward zero.
3. **Release (or go to full stick)** and the map resets to a straight line: 0 stick = 0 °/s, full stick = the budget. **Released stick holds zero pitch rate**, so the nose stays exactly where you left it, even at 60° AoA.
4. **Ease below the threshold point** and it stays in rate mode until the jet has actually come back down to the threshold. Then rate and AoA command are blended over the next 4°.
5. **Floating hold point.** If the jet drifts back down to the threshold while your stick is still past the threshold point, that stick position becomes the new "threshold" point. Easing from there glides you back onto the 72°-per-stick line with a soft 2° knee, never a jump.
6. **Easing before the crossing** keeps you in AoA command. So a quick 0.7 pull, eased to 0.45 before the nose reaches 36°, simply settles at about 32°.

Bench example at 300 km/h, afterburner: a held 0.7 stick commands about 48°, crosses 36° at 32 °/s, and from there holds 32 °/s until the AoA limit catches it.

The HUD mod shows **RATE n°/s** while this mode is active.

The push side mirrors it

Pushing uses the same scale: −36° at half forward stick, full push = the −45° limit, and a threshold that drops from −36° to −25° as the wing's negative lift falls from 2 g to 1 g. Above about 250 km/h the −3.5 G limit stops a push long before −36°, so this only matters at low speed, like a push-over at the top of a zoom climb.

Step 5: command shaping (how fast the command builds)

A raw stick step is not sent straight to the jet. A shaping filter decides how quickly the command grows, and **its speed depends on how big the request is**:

- **AoA:** tiny steps build at a rate of 1.5 per second; steps of 15° or more build at 10 per second, with a smooth curve in between.
- **G:** 1.5 per second for a small step up to 10 per second for 3 G or more.
- **Very small steps** also ease in over their first 0.25 s.
- **A big request keeps its speed** all the way in, because the filter remembers the step size.

The result: fine tracking corrections are silky (holding 26° with stick noise, AoA wobble dropped from 0.096° to 0.021°), while big pulls still snap. A 1 G step now takes about 1.4 s to reach 90 %, on purpose. The idea is the same as the self-adjusting pre-filter in the T-50's control law.

Step 6: arriving at the target without overshoot ("capture")

As the jet approaches the commanded AoA or G, the FCS calculates **how much nose-down braking it actually has at that AoA**, counting the stabs and the nozzles at the thrust the engines are making right now. It then slows the nose so it arrives exactly on target.

- **Overshoot (bench):** afterburner 0.0° mean / 1.5° worst; military power 0.0° / 1.6°.
- **Holding accuracy:** steady holds up to 35° within about 0.2°; riding the stick in small steps settles within ±0.3° in about 0.8 s.
- **Anti-windup:** when a limiter is holding the command back, the loop's memory (integrator) bleeds off instead of building up, so there is no slow creep afterwards.
- **At very high speed** the G is captured the same way: loading and unloading don't overshoot or bounce.

Step 7: letting go

- **Released stick = zero pitch-rate hold.** The FCS keeps the nose still while AoA or G bleeds off naturally.
- **Easing the stick** lowers the pitch rate smoothly instead of shoving the nose down.
- **In the G regime** the pitch rate is taken off over about 0.3 s, not in one step.

Pitch while rolling

Rolling hard disturbs pitch in two ways: the flaperons make some pitch moment when they roll, and a rolling jet wants to pitch up from its own inertia (inertial coupling). The FCS handles both:

- **Predicting the flaperons' pitch moment one servo step ahead** (not all at once), so the pitch correction arrives with the disturbance, not before it.
- **Stabs and nozzles share the work.** With only pitch stick in, pitch gets full authority. With lateral stick also in, the stabs are given to roll and yaw in proportion to the lateral stick (up to 70 %), and the nozzles carry what pitch they can. The FCS would rather limit pitch rate than slam the nozzles to full deflection.
- **Above about 150 m/s IAS in a low-AoA roll,** the stabs take the roll's pitch trim so the nozzles only move a few degrees.
- **Released stick gets pitch priority.** If you let go of pitch and roll, the pitch hold is not traded away for roll. Bench worst pitch dip in pull-release-roll at 450–600 km/h: −6.6 °/s (was −65 °/s).

Roll axis (stick left and right)

Lateral stick asks for a **roll rate around the flight path** (velocity-vector roll), not around the nose. Letting go holds zero roll rate. The maximum rate is whatever the jet can both reach and stop again at that moment.

Why roll around the flight path?

Imagine the jet at 30° AoA. If it rolls around its own nose, the nose sweeps sideways and the air starts hitting the side of the jet: sideslip. That kills the turn and can depart the jet. Rolling around the flight path instead keeps the nose on a cone around the direction of travel, so sideslip stays near zero.

To do that, the FCS splits the roll command into two body rotations:

- **Body roll** = roll command × cos(AoA)
- **Body yaw** = roll command × sin(AoA)

At 0° AoA it is all roll. At 30° half of it is yaw. By about 36° and above, roll and yaw become effectively **one axis**: the rudders, stabs and flaperons all work together to swing the nose around the flight path, like the real F-22.

How fast can it roll?

The roll-rate ceiling is the **lowest** of these:

1. **Steady-state authority.** The FCS estimates, live, how much roll moment the surfaces can make and how much the air resists rolling (roll damping). Their ratio is the fastest rate the jet can sustain with full deflection.
2. **Stoppability.** The jet must be able to stop the rate within about 1 s using its present roll acceleration. A heavy store far out on a wing lowers this automatically, even when the load is symmetric.
3. **Yaw coordination.** At moderate and high AoA, a velocity-vector roll needs body yaw. Once the rudders run out, the roll rate is capped so sideslip stays under about 2°.
4. **An absolute ceiling** far above what the airframe reaches in practice.

Bench steady roll rates at 3 km, low AoA:

True airspeed	Roll rate
80 m/s (288 km/h)	117 °/s
150 m/s (540 km/h)	about 199 °/s
220 m/s (792 km/h)	258 °/s
300 m/s (1080 km/h)	295 °/s

Starting a roll

- **Uses only the authority left over** after fighting roll damping at the present rate, so the FCS never plans a roll that would need more than full deflection.
- **Keeps a reserve for stopping.** Half of the roll authority is planned for acceleration at normal speeds, rising to 80 % at very low speed (≤ 60 m/s IAS), where the air's damping is weak and the surfaces are far from their stops.

- **Waits for the rudders at moderate AoA.** The roll's starting acceleration is also capped by how fast the yaw axis can keep up, so the jet does not slip sideways at the start of a roll at 20–30° AoA.

Stopping a roll

- **Braking budget** = 80 % of the wing surfaces' authority plus the help from roll damping at the present rate. As the rate falls, the budget tapers, so the surfaces are near neutral just as the roll stops. No rebound.
- **A small lead** makes up for the lag in the loop at high speed, where damping is strong.
- **The braking curve is calculated one physics tick at a time (tuned at 50 Hz)**, so it lands exactly on zero instead of ending in a jerk.
- **Result (bench):** low-AoA stops in about 0.6–0.7 s at 300–400 km/h, and about 0.5 s at 500–800 km/h.

Rolling at high AoA

- **While AoA is still building post-stall,** the roll rate is held back (for example to about 25 °/s at 550 km/h while AoA climbs from 30° to 50°). A hard roll there adds nose-up inertial coupling that could push AoA past 65°. Once AoA settles, the full rate comes back (48 °/s in that example).
- **High-AoA braking floor.** Deep in the post-stall region the roll and yaw rate limits are sized so the jet can always brake them. The reference is how hard a pedal turn with full lateral stick at max AoA can be braked at about 270 km/h. Below about 210 km/h the margin grows smoothly to 1.5×.
- **It anticipates slowing down.** Speed bleeds fast in a post-stall corkscrew, and control power goes with it. The limits look at a filtered estimate of how fast IAS is falling and tighten *before* the authority is gone, not after.

Which surfaces roll the jet

- **Both flaperon pairs** do most of the work at low AoA.
- **The stabs** join in differentially, more so at low speed (below about 240 m/s IAS) where pitch needs them less.
- **The rudders** supply the coordinating yaw, not roll.
- **No roll thrust vectoring.**

Special cases

- **Asymmetric jet.** If the jet is lopsided (store on one wing, battle damage), rolls toward the heavy side are limited to 120 °/s, and each roll direction's limit uses only the authority left after trimming out the imbalance.
- **Flying backwards.** Past 100° AoA the roll command is flipped (released again below 80°), so the jet still rolls the way you push the stick from your seat.

Yaw axis (pedals)

With pedals released, the yaw axis works silently to keep sideslip at zero and to supply the body yaw a velocity-vector roll needs. Pressing a pedal asks for sideslip at low AoA and swings the nose around at high AoA.

What the pedals command

- **Below 36° AoA: sideslip.** Full pedal asks for 15° of sideslip. The FCS turns that into a yaw-rate target, and the allocator spreads the yaw demand over every surface that can make yaw.
- **Above 36° AoA:** roll and yaw are one combined axis. The pedal swings the nose around the flight path; full aft stick plus alternating pedal is the classic **falling leaf**.
- **Pedals released:** zero sideslip. The rudders coordinate every roll automatically.

How fast can it yaw?

Like roll, yaw rate is limited by what the jet can **stop** again:

- **Stoppable cap.** Above 15–25° AoA, body yaw rate is capped at 0.7 × the measured yaw authority (in °/s²), so the rotation can always be braked. This is speed-dependent in effect, but it comes from measured authority, not a table.
- **Low-speed boost.** Below 120 m/s IAS the cap (and its floor) is widened by 35 %, fading back to normal by 180 m/s. That is what gives a lively falling leaf.
- **Braking floor.** Deep post-stall, the yaw limit is referenced to how well the jet can brake a max-AoA pedal turn at about 270 km/h, with an extra margin growing to 1.5× below 210 km/h. It also tightens early when IAS is falling fast.
- **Design target:** 30–36 °/s in the falling leaf, snappy at moderate speed, with some extra braking lag accepted at the very slow end.

Bench, full aft stick with alternating full pedal at 65° AoA:

IAS	Max yaw rate
300 km/h	52 °/s
270 km/h	46 °/s
240 km/h	39 °/s
210 km/h	31 °/s
180 km/h	19 °/s
150 km/h	11 °/s

In a held max-AoA turn (full aft plus full lateral stick) entered at 280 km/h, the yaw rate holds 26–28 °/s all the way down from 210 to 158 km/h.

The rudder problem: canted fins roll the jet

The F-22E's fins lean outward. So every rudder input also makes a **roll the opposite way**, and because the jet resists roll far less than yaw (roll inertia is about a third of yaw inertia), that roll is about **twice as strong as the yaw** even at 0° AoA. The FCS handles this in three ways:

1. **Cancel it.** The allocator knows the rudders' roll and uses the stabs and flaperons to cancel it. That is why you see the horizontal surfaces move when you press a pedal.
2. **Don't let surfaces fight.** While a pedal is in (and for 1 s after), the allocator is penalised for rolling one group of horizontal surfaces against another just to scrape extra yaw from drag. Without that rule, a pedal input with the rudders saturated made the stabs and flaperons split wildly.
3. **Fade the rudders at high AoA.** From 35° to 50° AoA the rudders' normal yaw job fades out, because there the inverse roll starts to fight the flight-path roll.

The rudders' second life: sideslip killers above 55°

Above about 55° AoA the same inverse roll becomes useful. At that attitude, rolling the body moves sideslip directly, and the rudders make more than three times as much sideslip correction as flight-path roll. Meanwhile the stabs and flaperons lose their sideslip authority near 60° and reverse past about 62°. So above the fade, the rudders get a **limited sideslip-correction range**:

- **One direction only:** they may move only the way that reduces sideslip error.

- **Scaled by the error:** nothing below 0.5° of error, full at 3°.
- **Yaw-capped:** travel is limited so the rudders add no more than 2 °/s² of body yaw. Their yaw per degree falls with AoA, so the allowed travel opens up: about 10° at 55° AoA, full travel near 65–70°.
- **Off** above 100° AoA (reversed flow) and on the ground.

Effect on the bench, falling leaf at 120 km/h: mean yaw rate 19.6 → 25.0 °/s and peak sideslip 11.3° → 5.8°.

Smoothness details

- **Yaw has priority over roll** in the allocator at low AoA (weight 8 below about 20°), so rolls stay coordinated instead of sliding.
- **Yaw feed-forward** (anticipating the yaw a roll will need) runs at 50 % below 200 m/s IAS so the rudders don't slam at every quick roll reversal, and at 100 % at higher speed and above about 28° AoA, where the body yaw *is* the flight-path roll.

Surface by surface

No surface is hard-wired to one axis. Each one does whatever mix of pitch, roll and yaw the allocator finds cheapest at that instant, using its real, stall-aware aerodynamics. Below is what each one typically ends up doing.

Stabilators

- **Pitch (moved together):** the main pitch control at medium and high speed.
- **Roll (moved opposite):** a strong helper, especially below about 240 m/s IAS where pitch needs less of them.
- **Yaw and sideslip:** at moderate-to-high AoA their drag and side-force help swing the nose, until that authority collapses near 60° AoA.
- **Travel:** +35° trailing edge up / −30° trailing edge down. The FCS works out which way is "trailing edge up" for each stab from its own aerodynamics the first time the jet moves faster than about 10 m/s; until then both are held to ±30°.
- **Why only one stab seems to move at moderate AoA:** at about 29° jet AoA, the stab going trailing-edge-down is already at its own stall peak (local AoA around 34°). More deflection would just stall it. So the FCS rolls with the other stab, by making it give up lift instead.

- **Stuck-stab rescue:** if a stab ends up parked deep in stall doing nothing useful, the nozzles take over its pitch work so it can go back to rolling and yawing.
- **Sharing with the nozzles:** with pitch and lateral stick held together, up to 70 % of the stabs' authority goes to roll and yaw in proportion to the lateral stick; the nozzles carry the pitch.

Flaperons (inner and outer pairs)

- **Roll:** the primary roll surfaces at low and moderate AoA.
- **Flaps (both deflect trailing edge down for extra lift):** the inner pair follows a flap schedule; the outer pair droops to 60 % of the inner angle. Example: at 111 m/s and 13° AoA the inner ones sit at 14.6° and the outer at 10.5°; at 218 m/s, 5.1° and 3.1°.
- **Stall-capped:** the flap angle is limited by each flaperon's own local AoA, so it never droops into separated flow, and the schedule fades out by 30° jet AoA.
- **Roll comes first:** with full lateral stick the flap droop retracts to leave roll travel.
- **Load relief:** at high load they move trailing edge up to take stress off the wing.
- **Pitch side effect:** rolling flaperons make some pitch moment. The FCS predicts it one servo step ahead and cancels it with the stabs and nozzles.
- **Slat coupling:** the leading-edge device mechanically swings the flaperon 15° trailing edge down. The flap drive subtracts that, so the net angle is always what the schedule asked for.

Rudders

- **Yaw and coordination:** supply the body yaw a velocity-vector roll needs, and hold zero sideslip.
- **Pedal response:** the main sideslip effector at low AoA.
- **Inverse roll:** always cancelled by the horizontals (see the yaw section).
- **High AoA:** normal job fades from 35° to 50° AoA; above that they work only as limited, one-way sideslip correctors, useful to about 85°.
- **Ground:** a nose-wheel-steering helper from 10–25 m/s upward, and toed into each other for aero-braking.

Thrust-vectoring nozzles (pitch only)

- **Pitch power that doesn't depend on airspeed:** the key to controlling the nose at low speed and post-stall.

- **Unloading the stabs:** the design intent is that the nozzles take pitch work so the stabs are free for roll and yaw.
- **Strength follows thrust:** the FCS reads live thrust every frame. At idle they do little, so the low-thrust AoA cap kicks in (see Protections).
- **Game cut-off:** Nuclear Option switches thrust vectoring off above 450 m/s true airspeed (about 600–650 km/h IAS at 16 km). The FCS counts them as zero there.
- **Gentle at roll stop:** when a roll ends, pitch is handed back from the nozzles to the stabs at 5 °/s, so there is no pitch bump.
- **Visuals:** optionally drawn as two flaps, each hinged at its own root, opening and flaring with afterburner. This is animation only; physics are unchanged.

Slats (leading-edge devices)

- **In this game the slats' only effect is extra wing area** (about +4 m²). Their shape is animation.
- **Driven by AoA:** they start coming down at 2° AoA and are fully down by 20°, and stay down up to 70° (fading over the 10° below it).
- **Also by speed:** the stock schedule (deployed below 100 m/s, retracted above 130 m/s) still applies; whichever asks for more wins.
- **Retracted when unloaded or pushing negative.**
- **Fully down during ground aero-braking.**

Speedbrake

- **Fully manual.** Only the brake control deploys it, in the air or on the ground, at any throttle.
- **Idle throttle never deploys any brake.** The stock mod did this automatically; the addon removes that.
- **No split surfaces on the ground** unless you press the brake (then the ground aero-braking configuration takes over).

Protections and limiters

The protections exist so a pilot can yank and bank without thinking about departures. Most of them do not just react to a limit being crossed; they predict where the jet is going and act before it gets there.

AoA protection

- **The limit is +65° / −45°** (or lower when one of the caps below is active).
- **Normally the capture does the job.** The kinematic capture in the pitch law (braking with the authority actually available) stops the jet at the limit; on the bench a max-AoA pull holds 64.4–64.9°.
- **If the jet goes past anyway**, protection drives the nose back at up to the full 48 °/s, regardless of thrust. The harder you hold aft stick, the less it pushes; full aft stick cancels the push and simply holds.
- **Protection only ever limits in the nose-up direction.** If you push, ease or release at the limit, your command takes effect immediately. (Before 1.9.6 it could pin the jet at ~65.5° against a full push for seconds.)
- **The negative side mirrors it.**

G limiter

- **+9.5 / −3.5 G** normally, +12.5 G in MPO at high speed.
- **It looks ahead.** The limiter predicts the load 0.15 s ahead, stretching to 0.35 s when rolling faster than 150 °/s, because a coordinated rolling pull loads up faster.
- **Bench, rolling pulls with full aft + full roll (limit 9.5 G):** 9.44 G at 250 m/s, about 9.6 G at 400 m/s.

Asymmetric-load AoA cap

A heavy store on one wing makes a roll and yaw moment that grows with AoA. The FCS's inner loop trims that out on its own, because it works from measured rotation. What it cannot do is create authority that isn't there. So:

1. **Every 0.2 s** (spread over several frames so there is no stutter), the FCS sweeps the whole airframe through AoA −40° to +65° at the current speed, air density and thrust, using the same parts and airfoils the game uses.
2. **It measures the imbalance** at each AoA. No loadout table: anything lopsided, including battle damage, shows up the same way.
3. **It measures what the surfaces can still do** at each AoA, counting the rudders' fade and never counting their inverse roll as useful roll.
4. **It caps the AoA** where the imbalance would leave the pilot too little roll and yaw to fly with, and **remaps the stick** onto the smaller range, so full aft is still meaningful.

5. **It moves gently:** the cap tightens at 10 °/s, relaxes at 4 °/s, and recovers AoA from above the cap at 15 °/s. It anticipates the speed loss in manoeuvres like an inverted corkscrew rather than reacting late, and it is filtered so it never jitters with IAS.

Small imbalances that the surfaces can handle are left to the inner loop and never touch the cap.

Low-thrust AoA cap

At idle, the nozzles have almost no pitch power. If the jet is pulled high enough, the stabs alone may not have the nose-down authority to get it back. So the FCS caps AoA at the point where **nose-down authority runs out, minus a safety margin**, counting the stabs at full nose-down, the flaperons at their trim, and the nozzles at the thrust they have right now.

Throttle at 350 km/h	AoA cap
Zero thrust	23.4°
About 0.1 thrust-to-weight	25.9°
30 % throttle	33.6°
Afterburner	none (full 65°)

Bench, zero thrust, full pull then full push: before the cap the jet reached 80° and never recovered; with it, 24.6° and a clean recovery. The cap tightens in 0.3 s and relaxes in 1 s.

The same cap solves a high-altitude problem: above 450 m/s true airspeed the game freezes the nozzles, and the stabs alone cannot hold much above 20–25°. At 16 km and 600 km/h IAS, a full pull used to run away to 75°; now it stops at 23°.

Battle damage

- **Learned trim.** The FCS learns the roll moment of a damaged jet while it is not rolling, and holds that estimate during rolls.
- **Direction-aware roll limits.** Each roll direction gets only the authority left after that trim, so the jet can always stop a roll toward its bad side.
- **Bench, one wing removed, 160 test cases from 350 to 650 km/h:** 0 departures (before: 29), standing sideslip 0.3° (before: 2.5°).

Odd attitudes

- **Tail slides:** AoA is unwrapped through ±180°, so the FCS never mistakes a tail slide for a sudden negative-limit violation.
- **Flying backwards:** past 100° AoA the roll sense is flipped so the stick still rolls the way you expect.

On the ground

On the runway the FCS keeps the jet on a perfectly straight line with no pedal input, turns pedal into a controlled yaw rate, and steers out crosswind drift at touchdown. These laws apply to human pilots; AI jets keep the game's stock steering.

Pitch on the ground

- **Below 45 m/s (162 km/h):** the stick moves the surfaces directly, like a normal taxiing jet.
- **Rotation law:** above 45 m/s, once the main wheels are down and the nose wheel has been off the ground for 0.15 s, pitch goes closed-loop for a smooth rotation or de-rotation. It disarms 8 m/s below the arming speed.

Nose-wheel steering (fly-by-wire)

Pedal = yaw rate, not wheel angle.

- Full pedal asks for up to 15 °/s of yaw at taxi speed, limited to 2.5 m/s² of sideways acceleration at speed. That works out to about 5 °/s at 100 km/h and 2 °/s at 260 km/h.
- The command ramps in at 20 °/s², so there is no lurch.
- Below 1.5 m/s the pedal steers the wheel directly, so you can pivot.

Pedal released = hold the line.

1. The yaw rate is braked to zero.
2. Once it is below 0.7 °/s, the FCS remembers the current track and the line through the CG.
3. It then steers back onto that line if anything pushes the jet off it (crosswind, bumps), aiming at a point ahead (at least 25 m, or 4 s of travel) with at most 5° of correction.

How the wheel is steered: the FCS works out the wheel angle geometrically from the yaw rate it wants, then adds a correction loop. The correction is capped just above the tyre's grip limit, so it never skids the nose wheel.

The rudders help with the same yaw-rate loop, fading in between 10 and 25 m/s.

Touchdown drift correction

- While airborne on approach, the FCS tracks the ground track direction.
- **At touchdown** it captures that approach track and the touchdown point as "the line", so a crabbed landing is steered straight and any drift is brought back.
- **Wings level:** while rolling on the mains, the roll axis levels the wings instead of holding bank, because a crabbed tyre's side force can tip the jet onto one main gear.
- **Bounces** shorter than 1.5 s keep the same line.

Bench, flat runway:

Case	Stock steering	FBW steering
Takeoff roll 40 → 337 km/h, 8 m/s crosswind	83 m off the centreline	0.17 m
Touchdown 280 km/h, 12 m/s crosswind, 9° crab	ground loop	6.6 m peak, back to 2.0 m
Braking rollout from 250 km/h, 8 m/s crosswind	2.3 m	0.10 m

Ground aero-braking (landing rollout)

When **all three wheels are on the ground and the brake control is pressed**, the jet takes up the real F-22's rollout configuration:

Surface	Position
All four flaperons	Trailing edge up, 20°
Stabilators	Trailing edge up, 26° (nose held high for drag)
TVC nozzles	Up, 10°
Rudders	Split 25°, toed into each other (pure drag, no yaw)

Surface	Position
Slats	Fully down
Split speedbrake surfaces	Out (can be turned off in config)

It never triggers from throttle: closing the throttle deploys nothing. Your stick and pedal inputs still add on top of it.

Cheat sheet

Item	Value
AoA limits	+65° / −45°
G limits	+9.5 / −3.5 G (MPO: +12.5 / −3.5 G)
Pitch-rate budget	up to 48 °/s
AoA stick scale	72° per full stick; half stick = 36°
AoA → pitch-rate hand-over	36° (falls to 25° at low speed), both signs
Rate/AoA blend band	4°
Roll command	velocity-vector roll rate; released = zero rate
Pedal (below 36° AoA)	15° sideslip at full pedal
Falling-leaf yaw target	30–36 °/s
Rudder fade	35° → 50° AoA, then one-way sideslip correction (yaw ≤ 2 °/s²)
Stabilator travel	+35° TE up / −30° TE down
Flaperon travel (both pairs)	±25°
Rudder travel	±35°
TVC	±20°, pitch only, off above 450 m/s true airspeed
Servo speed / ramp	100 °/s / 0.06 s, all effectors
Slats	2° → 20° AoA, down to 70° AoA
Ground: rotation law arms	45 m/s

Item	Value
Ground: full-pedal yaw rate	15 °/s taxi, ⩽ 2.5 m/s² at speed
Loop rate	every physics tick (50 Hz at Unity's default step)

Honest caveats

- **Most numbers come from the bench, not from flying in the game.** The bench flies the real F-22E airframe data, but as one rigid body; the game's jet is 37 jointed parts. Trust the relative improvements more than the absolute numbers.
- **The bench assumes full fuel and a clean jet** unless a test says otherwise.
- **The newest features (the 1.9.x pitch-rate hand-over, push-side mirror, protection fix) are bench-verified only** so far.
- **If something flies wrong,** turn on telemetry in the developer build and attach the CSV it writes. That is how almost every fix in this addon was found.